

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Kalman Filter as Bayesian Estimator
Extended Kalman Filter
Unscented Kalman Filter

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Welcome to the lecture!

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- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

- **In the last lecture we looked at the "most basic" variant of the Kalman Filter.**
- **Today, we will**
  - interrelate the Kalman Filter to Bayesian estimators,
  - and look at nonlinear extensions –
    the Extended and the Unscented Kalman Filters.
- **In the next weeks, we will check out how to use Kalman Filters for Multi-Object Tracking (MOT).**

The Kalman Filter as a Bayesian Estimator

Bayes Optimal Filter

- **We have seen the Kalman Filter as a linear random process with Gaussian process and measurement noise:**

$$\hat{\mathbf{x}}_{t+1} = A \cdot \mathbf{x}_t + \mathbf{w}_t$$
$$\hat{\mathbf{z}}_t = H \cdot \mathbf{x}_t + \mathbf{v}_t$$

or in probabilistic terms

$$\hat{\mathbf{x}}_{t+1}|\mathbf{x}_t \sim p(\hat{\mathbf{x}}_{t+1}|\mathbf{x}_t) = \mathcal{N}(A\mathbf{x}_t, Q_t)$$
$$\hat{\mathbf{z}}_t|\mathbf{x}_t \sim p(\hat{\mathbf{z}}_t|\mathbf{x}_t) = \mathcal{N}(H\mathbf{x}_t, R_t)$$

- **We can see this as a specific case for a general nonlinear random process:**

$$\hat{\mathbf{x}}_{t+1} = f(\mathbf{x}_t, \mathbf{w}_t)$$
$$\hat{\mathbf{z}}_t = h(\mathbf{x}_t, \mathbf{v}_t)$$

or in probabilistic terms

$$\hat{\mathbf{x}}_{t+1}|\mathbf{x}_t \sim p(\hat{\mathbf{x}}_{t+1}|\mathbf{x}_t)$$
$$\hat{\mathbf{z}}_t|\mathbf{x}_t \sim p(\hat{\mathbf{z}}_t|\mathbf{x}_t)$$

- **How would an optimal Bayes filter look like?**
  - Find the Maximum A Posteriori estimate:

$$\hat{\mathbf{x}}_t = \underset{\mathbf{x}}{\operatorname{argmax}}\ p(\mathbf{x}_t|\mathbf{z}_t) = \underset{\mathbf{x}}{\operatorname{argmax}}\ \frac{p(\mathbf{z}_t|\mathbf{x}_t)p(\mathbf{x}_t)}{p(\mathbf{z}_t)}.$$

  - For symmetric distributions (like the normal distribution), the Minimum Mean Square Error and the Max A Posteriori Estimate are the same.

Bayes Optimal Filter

- **Since we do not have all measurements from the beginning of the universe to *t* available, we need to do the Max A Posteriori estimate $\hat{\mathbf{x}}_t = \underset{\mathbf{x}}{\operatorname{argmax}}\ p(\mathbf{x}_t|\mathbf{z}_t)$ recursively.**
- **We introduce the *Markov Assumption:***
  "A system's future state depends only on its current state, not on the sequence of events that preceded it".
- **For us that means that all past information is condensed into the state $\mathbf{x}_t$, or mathematically**

$$p(\mathbf{x}_t|\mathbf{z}_t) = \frac{p(\mathbf{z}_t|\mathbf{x}_t)p(\mathbf{x}_t)}{p(\mathbf{z}_t)} = \frac{p(\mathbf{z}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{z}_{t-1})}{p(\mathbf{z}_t|\mathbf{z}_{t-1})}$$

Prior taken from previous Posterior.

- **Now we can perform recursive iteration using the model functions.**

Bayes Optimal Filter

- **We want to determine the prior** $p(\mathbf{x}_t|\mathbf{z}_{t-1}) \overset{t\leftarrow t+1}{=} p(\mathbf{x}_{t+1}|\mathbf{z}_t).$
- **We remember that we can get rid of the conditional by integrating over it (*Marginalization*):**

$$p(x) = \int_{-\infty}^{\infty} p(x, y)dy$$

- **We can also resolve multiple conditional probabilities:**

$$p(x, y|z) = p(x|y) \cdot p(y|z)$$

- **With those two rules, we can determine**

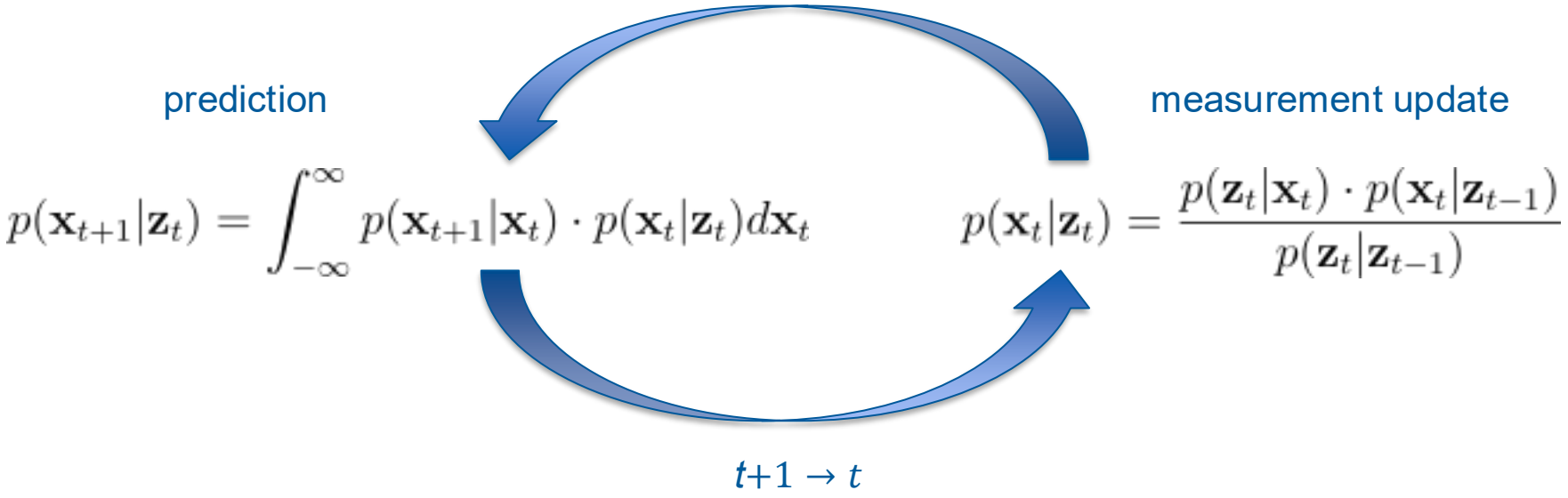
Transition Model

Prior from previous update

$$p(\mathbf{x}_{t+1}|\mathbf{z}_t) = \int_{-\infty}^{\infty} p(\mathbf{x}_{t+1}, \mathbf{x}_t|\mathbf{z}_t)d\mathbf{x}_t = \int_{-\infty}^{\infty} p(\mathbf{x}_{t+1}|\mathbf{x}_t) \cdot p(\mathbf{x}_t|\mathbf{z}_t)d\mathbf{x}_t$$

Bayesian Filter Recursion - Summary

- Bayesian Filter recursion:



- Let's apply these thoughts to our linear random process:

$$\mathbf{x}_{t+1} = F\mathbf{x}_t + \mathbf{w}_t,\ \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q_t)$$
$$\mathbf{z}_t = H\mathbf{x}_t + \mathbf{v}_t,\ \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, R).$$

Note: Using F instead of A from now on for the state transition to get closer in terminology to the general case of $\mathbf{x}_{t+1} = f(\mathbf{x}_t)$.

Bayesian Filter Recursion and the Kalman-Filter - 1

- We know that our prediction rule is

$$p(\mathbf{x}_{t+1}|\mathbf{z}_t) = \int_{-\infty}^{\infty} p(\mathbf{x}_{t+1}|\mathbf{x}_t) \cdot p(\mathbf{x}_t|\mathbf{z}_t) d\mathbf{x}_t$$

$$\mathbf{x}_{t+1} = F\mathbf{x}_t + \mathbf{w}_t, \;\; \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q_t)$$
$$\Rightarrow p(\mathbf{x}_{t+1}|\mathbf{x}_t) = \mathcal{N}(F\mathbf{x}_t, Q_t)$$

Transition Model

$$p(\mathbf{x}_t|\mathbf{z}_t) = \mathcal{N}(\hat{\mathbf{x}}_t, P_t)$$

Previous Measurement / Update

$$p(\mathbf{x}_{t+1}|\mathbf{z}_t) = \int_{-\infty}^{\infty} \mathcal{N}(F\mathbf{x}_t, Q) \cdot \mathcal{N}(\hat{\mathbf{x}}_t, P_t) d\mathbf{x}_t.$$

Bayesian Filter Recursion and the Kalman-Filter - 2

- Prediction

$$p(\mathbf{x}_{t+1}|\mathbf{z}_t) = \int_{-\infty}^{\infty} \mathcal{N}(F\mathbf{x}_t, Q) \cdot \mathcal{N}(\hat{\mathbf{x}}_t, P_t) d\mathbf{x}_t.$$

- It can be shown (for a derivation see [1]) that for two Gaussian distributions

$$\mathcal{N}(A\mathbf{x} + \mathbf{b}, Q), \ \mathcal{N}(\mathbf{m}, P)$$

  the integral of the product is

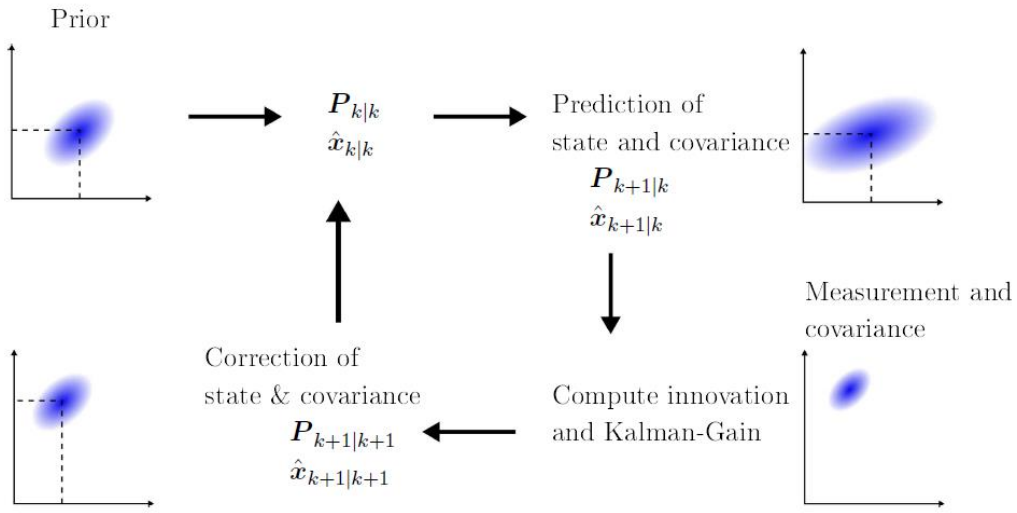
$$\int \mathcal{N}(A\mathbf{x} + \mathbf{b}, Q) \cdot \mathcal{N}(\mathbf{m}, P) d\mathbf{x} = \mathcal{N}(A\mathbf{m} + \mathbf{b}, APA^\top + Q).$$

- Using this trick, we obtain $p(\mathbf{x}_{t+1}|\mathbf{z}_t) = N(F\hat{\mathbf{x}}_t, FP_tF^\top + Q_k).$

$$\hat{\mathbf{x}}_{t+1} = F\hat{\mathbf{x}}_t$$
$$P_{t+1} = FP_tF^\top + Q_t$$

Standard Kalman prediction equations

- I'll spare you the derivation of the update equations...



Iterations of the Kalman-Filter

Kalman-Filter – Summary

- Prediction step (state estimate and covariance)

$$\hat{\mathbf{x}}_{t+1} = F\hat{\mathbf{x}}_k, \;\; P_{t+1} = FP_tF + Q_t$$

- Correction of state and covariance
- Compute the Kalman-Gain (we called P_{t+1} P'_t before)

$$K_{t+1} = P_{t+1}H^\top (HP_{t+1}H^\top + R)^{-1}$$

- Correction of state and covariance

$$\hat{\mathbf{x}}_t = \mathbf{x}'_t + K_t \cdot (\mathbf{z}_t - H\mathbf{x}'_t)$$
$$P_t = (I - K_tH) \cdot P'_t.$$

- A-Posteriori covariance of the measurement: $P_{zz} = R_t + HP_tH^\top$
- Cross-covariance of estimate and measurement: $P_{xz} = P_tH^\top$
- **Please note: The Kalman Gain is the product $K_{t+1} = P_{xz} \cdot {P_{zz}}^{-1}$ which will be an insight used when deriving the unscented Kalman Filter.**

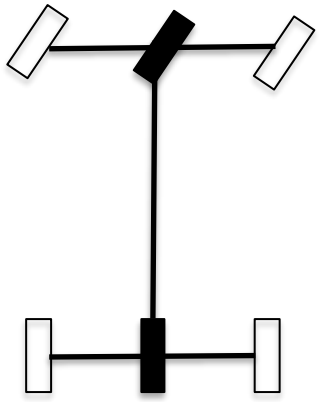
Modeling Nonlinear Systems

or - why a linear Kalman Filter is not enough

- **The regular Kalman Filter is a powerful tool – but it is limited to linear processes, i.e. processes that can be described using** $\mathbf{x}_{t+1} = F\mathbf{x}_t + \mathbf{b}.$
- **In many cases, this works well enough; however it fails for nonlinear processes.**
- **The Kalman Filter family tree is a forest; but the two most basic extensions are**
  - The Extended Kalman Filter:
    - Linearizes by approximating the non-linear system around the current state estimate using Jacobians.
    - Then applies the standard predict/update cycle of the linear Kalman Filter to this approximation.
    - Widely used, however not too acurate due to the approximation used.
  - The Unscented Kalman Filter:
    - Uses a deterministic sampling approach called the **unscented transform** instead of linearization to address the inaccuracies of the Extended Kalman Filter.
    - More computationally intensive, so less often used.

Nonlinear State Estimation Example: Simple Vehicle Model

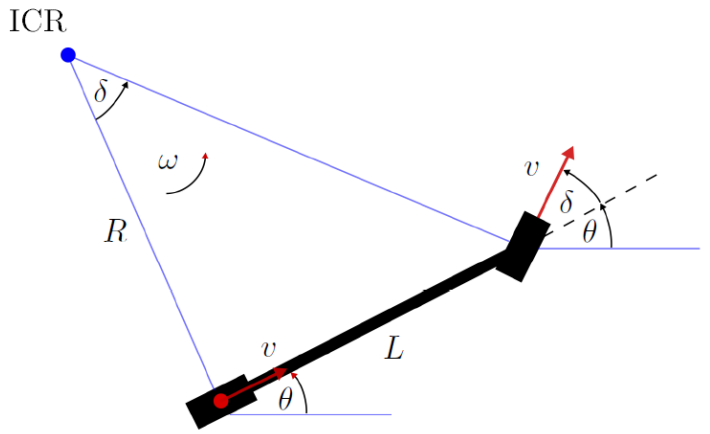
- Most multi-object trackers modeling vehicular motion use the Single-Track/Bicycle Model, although it is not a very precise model:
  - It neglects the roll movement of the vehicle
  - It neglects tire effects including suspension
  - It cannot model lateral accelerations in high-speed maneuvering situations



Single-Track Model of a car

Task: Derive a state-space model (reference: rear axis)

- States are position x, y, heading angle θ, steering angle δ
- Input u is the steering rate
- Use instantaneous center of rotation (ICR)
- Also take into account
  - Geometry, length L is known
  - Speed v is known



Single-Track Model: reference point is rear-axis

Single track model (reference: rear axis)

- For the position we have that

$$\dot{x} = v \cdot \cos(\theta)$$
$$\dot{y} = v \cdot \sin(\theta)$$

- The derivative of the heading angle reads

$$\dot{\theta} = \omega = \frac{v}{R}$$

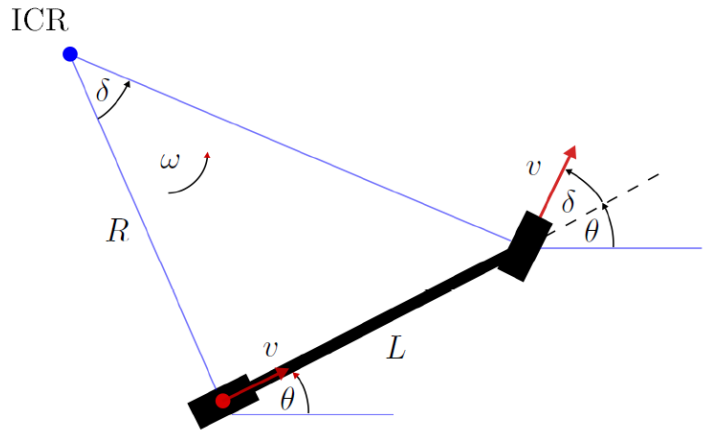
- The radius R can be inferred from geometry:

$$\tan(\delta) = \frac{L}{R} \quad \Leftrightarrow R = \frac{L}{\tan(\delta)}$$

$$\dot{\theta} = v \cdot \frac{\tan(\delta)}{L}$$

- The steering rate is the input, so

$$\dot{\delta} = u_{steer}$$



Single-Track Model: reference point is rear-axis

$$\dot{x} = v \cdot \cos(\theta)$$
$$\dot{y} = v \cdot \sin(\theta)$$
$$\dot{\theta} = v \cdot \frac{\tan(\delta)}{L}$$
$$\dot{\delta} = u_{steer}$$

Literature / Links

1. J. Duda: ***Gaussian Auto-Encoder***, https://arxiv.org/pdf/1811.04751v1
2. K.Chadaporn: ***Simple Example of Applying Extended Kalman Filter***, https://www.researchgate.net/profile/Keatmanee-Chadaporn/publication/273381901_Simple_Example_of_Applying_Extended_Kalman_Filter/links/54ffc8960cf2672e22489e9c/Simple-Example-of-Applying-Extended-Kalman-Filter.pdf



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Thank you very much!